A Rare Blend of Einsteinian Assumptions and Utter Batsh*t Insanity

Assumption Stress-Testing in Relativity, Spacetime, and Speculative Foundations of Physics

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Abstract

Foundational physics connects mathematical models to observations through assumptions about idealisation, dynamics, measurement, and inference. This paper proposes *Assumption Stress-Testing* (AST), a structured workflow for recording and examining these dependencies. The workflow tracks assumptions, mathematical structure, dynamics, observables, and empirical constraints, distinguishing mathematical consequences from physical interpretation and empirical support.

A worked example uses a phenomenological modified dispersion relation to derive a group-velocity expression and compare it with a perturbative approximation. The calculation is reproducible, but it is illustrative rather than an observational test. The paper also discusses the conformal ambiguity relevant to reconstructing spacetime geometry from causal structure and the schematic role of effective actions.

No claim is made that general relativity is false, Lorentz symmetry is violated, spacetime emergence has been established, or AST is a novel method in the strong priority sense. Its distinct contribution remains a hypothesis to be tested against existing practices in model checking, sensitivity analysis, and theory-to-observation inference. A reproducible implementation accompanies the manuscript.

Keywords: foundations of physics; assumption analysis; modified dispersion relations; Lorentz invariance; emergent spacetime; reproducibility; model testing

1 Introduction

Physical theories are not tested by equations in isolation. A theoretical conclusion typically depends on a model, auxiliary assumptions, a mathematical derivation, a map from theoretical quantities to observables, and an inferential procedure for comparing predictions with data. Errors or hidden choices at any link can change what a calculation means.

This paper proposes *Assumption Stress-Testing* (AST) as a compact workflow for recording and examining these links. The aim is methodological rather than revolutionary: to make the dependency structure of an argument inspectable, identify where a conclusion is conditional, and prevent a mathematical result from being presented as stronger physical evidence than it is.

The central question is not whether assumptions can be questioned. That practice is already present in mathematical physics, philosophy of science, uncertainty analysis, and model criticism. The question is whether a particular structured ledger and workflow can improve clarity, error detection, reproducibility, or comparison across models. That is a methodological question to be investigated, not an established result of this paper.

1.1 Scope and claims

This paper makes four limited contributions:

1. It defines a proposed representation of a theory-to-observation argument.
2. It provides an explicit audit sequence and reporting categories.
3. It demonstrates the sequence on a phenomenological modified dispersion relation.
4. It identifies tests needed to assess whether the workflow adds value beyond existing methods.

The paper does not introduce a new fundamental physical theory, derive a replacement for general relativity, provide new observational data, or establish the emergence of spacetime. The worked example is deliberately modest: it tests whether mathematical steps and inferential limits are stated correctly.

2 The AST Framework

2.1 Representation

Represent a theory-to-observation argument by

$$\mathcal{T} = (\mathcal{A}, \mathcal{M}, \mathcal{D}, \mathcal{O}, \mathcal{E}), \tag{1}$$

where:

- $\mathcal{A}$ is the set of assumptions and idealisations;
- $\mathcal{M}$ is the mathematical model or structure;
- $\mathcal{D}$ is the specified dynamics or rule of evolution;
- $\mathcal{O}$ is the mapping from theoretical quantities to potential observables;
- $\mathcal{E}$ is the empirical and inferential procedure, including data, uncertainty, and decision criteria.

This tuple is a bookkeeping proposal, not a theorem that every scientific argument must fit into exactly five categories. Real analyses may require additional layers, feedback between stages, or multiple competing models.

2.2 Audit sequence

For a specified claim, the proposed workflow is:

1. **State the claim.** Write the conclusion in a form that could in principle be checked.
2. **List assumptions.** Separate stipulated model choices, mathematical conditions, physical interpretations, and measurement assumptions.
3. **Trace derivations.** Identify the operations connecting assumptions to equations and the conditions under which those operations are valid.
4. **Map to observables.** Specify how a theoretical quantity would affect a measured quantity, including nuisance parameters and uncertainty.
5. **Compare alternatives.** Ask whether a competing model or parameter combination could produce observationally indistinguishable results.
6. **Classify the outcome.** Report what is mathematically established, empirically constrained, excluded under stated assumptions, not excluded, or not distinguishable with the available evidence.

7. **Record reproducibility.** Preserve equations, parameter values, data provenance, software versions, and executable steps.

The outcome labels are not interchangeable. Failure to exclude a model is not confirmation of it. A mathematical consequence is not an empirical finding. A bound on parameters within one model is not a universal bound on every theory sharing a broad label.

2.3 Assumption ledger

A minimal ledger should include:

Category	Question	Required record
Model	What has been stipulated?	Equations, parameters, domain
Mathematical	Which steps are valid?	Derivation and conditions
Physical	What do symbols represent?	Interpretation and regime
Measurement	What could be observed?	Observable map and uncertainties
Inference	What conclusion follows?	Statistical criterion and alternatives
Reproducibility	Can the result be checked?	Data, code, environment, commands

3 Worked Example: A Modified Dispersion Relation

3.1 Purpose and status

Modified dispersion relations have been studied as phenomenological ways to parameterise possible departures from conventional relativistic kinematics. Their theoretical interpretation and observational implications are discussed in the existing literature, including Mattingly's review [3]. The example below does not claim a new dispersion relation or new observational constraint. It demonstrates the distinction between a stipulated model, a derived consequence, and a physical or empirical claim.

3.2 Model specification

Use natural units $\hbar = c = 1$ and consider the massless ansatz

$$E^2 = p^2 + \eta \frac{p^4}{M^2}, \tag{2}$$

where E is energy, $p > 0$ is the magnitude of momentum, $M > 0$ is an energy scale, and η is dimensionless. Equation (2) is stipulated phenomenologically; it is not derived here from a fundamental theory.

Define

$$x = \eta \frac{p^2}{M^2}. \tag{3}$$

The positive-energy branch is

$$E(p) = p\sqrt{1 + x}. \tag{4}$$

For this branch to be real, $1 + x > 0$. The expansion below additionally assumes $|x| \ll 1$.

3.3 Group velocity

If the relevant wave-packet quantity is defined by $v_g = dE/dp$, then differentiation of Equation (4), using $dx/dp = 2x/p$, gives

$$v_g = \frac{d}{dp}\left[p(1+x)^{1/2}\right] \tag{5}$$

$$= \sqrt{1+x} + \frac{x}{\sqrt{1+x}} \tag{6}$$

$$= \frac{1+2x}{\sqrt{1+x}}. \tag{7}$$

Expanding around $x = 0$,

$$v_g = 1 + \frac{3}{2}x - \frac{5}{8}x^2 + O(x^3), \tag{8}$$

so the leading-order correction is

$$v_g - 1 \approx \frac{3}{2}\eta\frac{p^2}{M^2}. \tag{9}$$

These are mathematical consequences of the chosen ansatz and definition of group velocity. Their physical interpretation depends on the underlying propagation theory. Group velocity alone does not establish a signal velocity, a causal structure, or a violation of causality.

3.4 Numerical consistency check

The accompanying Python program evaluates the exact expression and first- and second-order approximations at selected dimensionless values of x. These values demonstrate numerical consistency; they are not experimental data.

x	Exact v_g	First order	Second order
0	1.0000000	1.0000000	1.0000000
10^{-6}	1.0000015	1.0000015	1.0000015
10^{-3}	1.0014994	1.0015000	1.0014994
10^{-2}	1.0149379	1.0150000	1.0149375

The approximation becomes more accurate as $|x|$ decreases in the displayed regime. This check validates the algebraic expansion against the exact expression; it does not validate the physical model.

3.5 AST ledger for the example

Stage	**Statement**	**Status**
Model assumption	Massless dispersion relation with a p^4/M^2 correction	Stipulated ansatz
Mathematical condition	$1+x > 0$ for real positive-branch energy; $\lvert x\rvert \ll 1$ for the expansion	Explicit validity conditions
Derived consequence	$v_g = (1+2x)/\sqrt{1+x}$	Follows from differentiation
Approximation	$v_g = 1 + 3x/2 + O(x^2)$	Valid for small $\lvert x\rvert$
Physical interpretation	Group velocity is relevant to the propagation question	Requires a specified underlying theory

Observation	An experiment constrains model parameters	Not performed in this paper
Inference	The ansatz is supported or excluded by data	Not established by this calculation

3.6 What an observational test would require

An observational analysis would need, at minimum:

1. a specified physical framework in which the dispersion relation applies;
2. an observable and a derived mapping from model parameters to that observable;
3. a data set with documented provenance and measurement uncertainties;
4. a treatment of nuisance parameters, selection effects, and competing explanations;
5. a statistical procedure specified before interpreting the result;
6. a comparison with published constraints and independent reproduction.

No numerical bound on η or M follows from the illustrative calculation alone.

4 Causal Structure and Spacetime Reconstruction

A second motivation for explicit assumption tracking is the relation between causal structure and spacetime geometry. Under suitable assumptions, causal information can constrain the conformal class of a Lorentzian metric rather than fixing the full metric uniquely. In broad terms, conformally related metrics can share light-cone structure while differing by a position-dependent scale factor. The exact reconstruction statement depends on the hypotheses and mathematical setting; it must not be treated as a universal claim without those conditions.

Consequently, a proposal to reconstruct spacetime from causal or relational data must state what additional information fixes the remaining freedom. Depending on the framework, this may involve volume information, a measure, dynamical equations, boundary conditions, or other structure. The existence of this reconstruction issue does not itself demonstrate that spacetime is emergent, nor does it identify a unique emergence mechanism.

An AST ledger for a reconstruction proposal should distinguish:

- input structure (causal order, adjacency, correlations, or other data);
- reconstruction map and its domain;
- uniqueness and equivalence claims;
- assumptions used to recover dimensional, metric, or dynamical information;
- observables that could discriminate the reconstructed model from alternatives.

A complete worked reconstruction is outside the scope of the present illustrative study.

5 Effective Dynamics and the Role of an Action

An effective action may organise the dynamics of a model, for example through an Einstein–Hilbert-type term,

$$S_{\mathrm{EH}} = \frac{1}{16\pi G} \int d^4x \, \sqrt{-g} \, (R - 2\Lambda), \tag{10}$$

in units where $c = 1$, with the usual qualifications about conventions, boundary terms, and matter contributions. Writing such a term does not derive it from a microscopic theory. Nor does it show that a proposed microscopic model produces general relativity in an appropriate limit.

For an emergent-spacetime proposal, deriving an effective action would require a specified microscopic or statistical model, a controlled coarse-graining or limiting procedure, and an account of the degrees of freedom and approximations retained. The AST workflow would record these as assumptions and derivation steps rather than treating the appearance of a familiar action as proof of emergence.

6 Relation to Existing Methods and Novelty

The general practice of exposing assumptions, checking limiting cases, studying parameter sensitivity, comparing competing models, and linking theories to observations is well established across physics and philosophy of science. The present paper does not claim priority for these broad activities. Mattingly [3] reviews phenomenological Lorentz-violation tests, while Will [4] surveys experimental tests of general relativity. These are relevant background sources, not proof that the proposed AST notation is novel.

To establish a distinct contribution, future work should compare AST with specific existing protocols for model verification, sensitivity analysis, uncertainty quantification, and theory-data comparison. The comparison should define measurable criteria such as:

- the number of hidden assumptions identified by each approach;
- detection rate for seeded derivation or inference errors;
- completeness of reproducibility records;
- time and effort required for independent reproduction;
- agreement between independent reviewers.

A useful evaluation could present reviewers with matched cases, some containing deliberately seeded omissions or invalid inferences, and assess whether the AST ledger improves detection compared with a baseline checklist. Such a study would need a pre-specified protocol, transparent scoring, and appropriate caution about sample size and reviewer effects.

Until this comparison is performed, AST should be described as a proposed organising workflow rather than a validated new methodology.

7 Limitations

This manuscript has several limitations:

1. The AST framework is conceptual and has not been benchmarked against existing methods.
2. The dispersion-relation example is a mathematical demonstration, not a fit to observational data.

3. No new experimental or observational result is reported.

4. The discussion of spacetime reconstruction is conceptual and does not provide a complete reconstruction theorem or model.

5. The effective-action discussion is schematic and does not derive an action from microscopic dynamics.

6. The bibliography is a starting point, not an exhaustive systematic review.

These limitations constrain the strength of the conclusions. They do not invalidate the algebraic example, but they prevent the paper from supporting claims of empirical confirmation, fundamental-theory replacement, or established methodological novelty.

8 Conclusion

This paper has proposed Assumption Stress-Testing as a structured way to document how assumptions feed into mathematical derivations, physical interpretations, observables, and empirical inferences. A modified dispersion relation served as a reproducible worked example: the group-velocity expression and its perturbative expansion follow from the stated ansatz, while observational implications remain conditional on additional physical and empirical structure.

The principal result is therefore methodological and limited: the example demonstrates how explicit bookkeeping can separate a stipulated model from its mathematical consequences and from claims that would require data. Whether the AST workflow provides a measurable advantage over established approaches remains an open question. Its next scientific test is a controlled comparison against existing methods, not a stronger rhetorical claim.

Reproducibility

The accompanying script `ast_dispersion_audit.py` evaluates the exact and approximate expressions, checks domain conditions, compares Taylor approximations, and writes a CSV table. Tests can be run using Python's standard-library `unittest` framework. The example requires no external scientific Python packages. Commands and expected behaviour are provided in the accompanying README.

References

[1] A. Einstein, "On the Electrodynamics of Moving Bodies," *Annalen der Physik*, 17, 891–921 (1905).

[2] A. Einstein, "The Foundation of the General Theory of Relativity," *Annalen der Physik*, 49, 769–822 (1916).

[3] D. Mattingly, "Modern Tests of Lorentz Invariance," *Living Reviews in Relativity*, 8, 5 (2005). https://doi.org/10.12942/lrr-2005-5.

[4] C. M. Will, "The Confrontation between General Relativity and Experiment," *Living Reviews in Relativity*, 17, 4 (2014). https://doi.org/10.12942/lrr-2014-4.

[5] T. Jacobson, S. Liberati, and D. Mattingly, "Lorentz Violation at High Energy: Concepts, Phenomena, and Astrophysical Constraints," *Annals of Physics*, 321(1), 150–196 (2006). https://doi.org/10.1016/j.aop.2005.06.004.

[6] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin, "Space-Time as a Causal Set," *Physical Review Letters*, 59, 521–524 (1987). https://doi.org/10.1103/PhysRevLett.59.521.

[7] R. M. Wald, *General Relativity*, University of Chicago Press (1984).